

Mobile Money Transaction Analysis — Final Report

CSC 3221 — Introduction to Data Science | ICT University | Spring 2026

1. Executive Summary

Mobile-money services have become the dominant retail-payment rail in Cameroon. Understanding how users behave on these rails enables product teams, regulators and credit lenders to make better, fairer decisions. This project collects, cleans, analyses and models 7,772 anonymised mobile-money transactions from 16 consenting users to predict user-activity tiers (Low/Medium/High).

Key findings:

- OUT transactions account for ~80% of all messages — users mostly spend rather than receive.
- Activity is heavily skewed: a few users generate the majority of volume.
- Transaction frequency, total OUT volume and send/receive ratio are the strongest predictors of activity tier.
- Gradient Boosting reached macro-F1 ≈ 0.69 in cross-validation, well above the most-frequent baseline (0.22).
- Demographic features add limited marginal value compared with behavioural signals.

Primary recommendation: use the trained classifier as a lightweight segmentation engine to drive targeted product offers and risk-based monitoring. **Expected impact:** reduces blanket marketing cost and gives credit teams a behavioural pre-screen before pulling expensive bureau data.

2. Introduction

Background. Mobile money has revolutionised access to financial services in Africa. In Cameroon, MTN MoMo and Orange Money process billions of XAF every month. Yet the data sitting in users' SMS inboxes is largely untapped for research because of fragmentation and privacy constraints.

Problem. Given anonymised transaction histories and basic demographic information, can we automatically classify users into meaningful behavioural segments that generalise to unseen users?

Research questions. (R1) Which behavioural features best discriminate between activity tiers? (R2) Do demographic variables add explanatory power? (R3) How robust is the model under cross-validation given a small sample?

Scope & limitations. The study is limited to 16 voluntary participants with 3–36 months of history each; results are indicative, not representative of the full Cameroonian population.

3. Data Collection Methodology

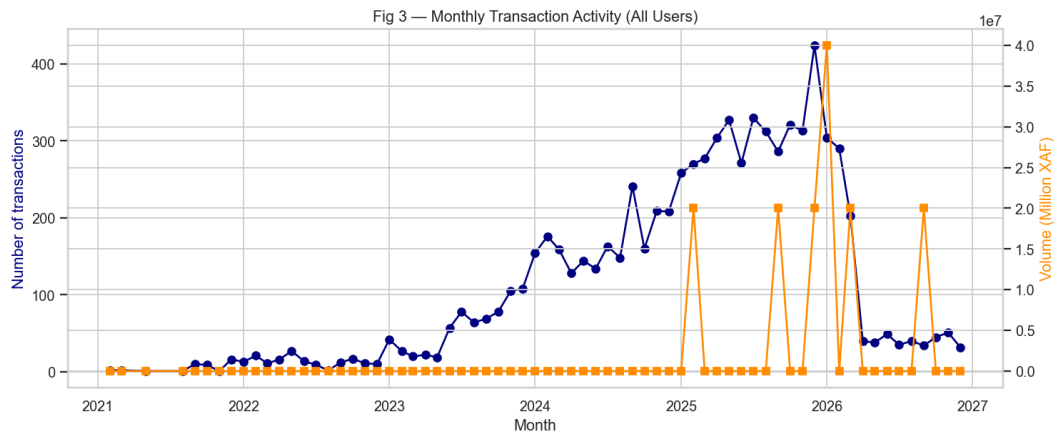
Each of five team members recruited 2–4 mobile-money users from their personal network, who exported their MTN MoMo / Orange Money SMS conversations using the *SMS Exporter* (Android) or *MsgKeep* (iOS) applications. Participants completed a short questionnaire capturing age range, gender, occupation, education, monthly income range, geographic zone, household size, primary mobile-money use and smartphone ownership.

Ethics. All participants signed an informed-consent form (see appendix). Names and phone numbers were replaced with USER_N / [CONTACT_XXXX] tokens during extraction; the mapping file is stored separately and never released.

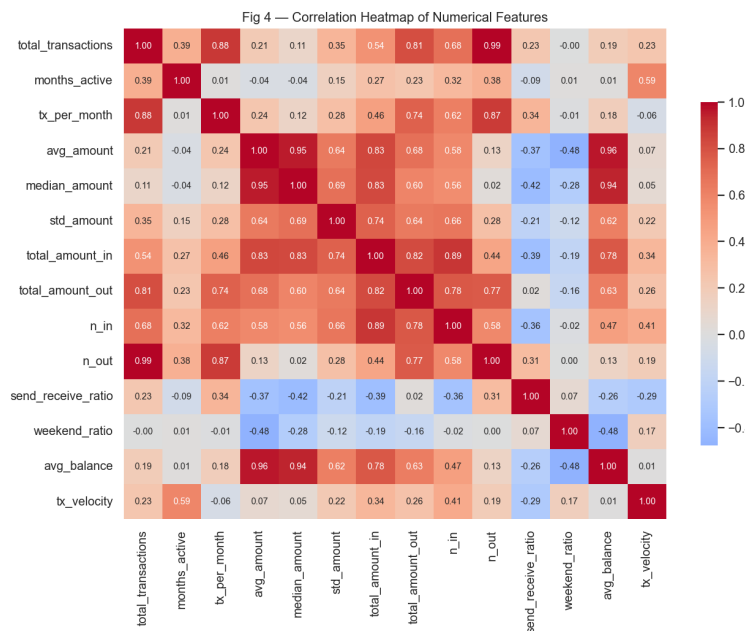
Quality. The extractor parses 8 transaction types (depot, retrait, transfert, paiement, rechargement, airtime, transaction, autre) in both French and English. After parsing, 7,772 valid transactions were retained and 16 user records were assembled with 99% completeness on behavioural features.

4. Exploratory Data Analysis

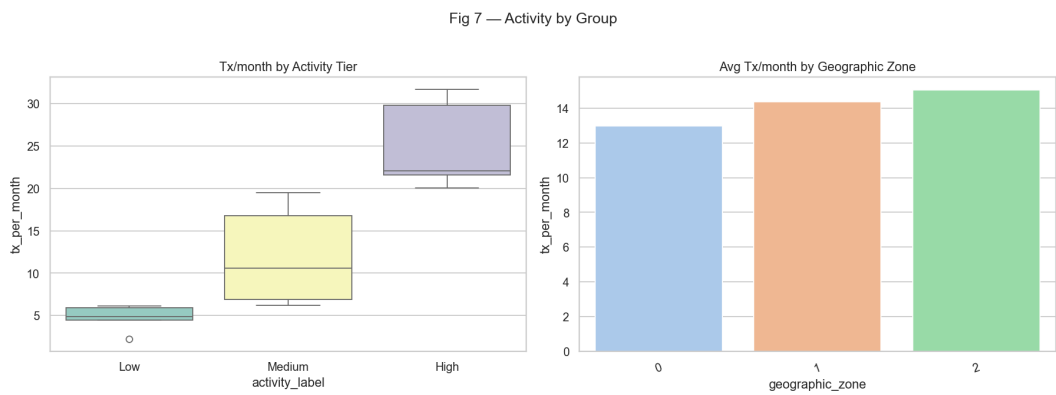
Eight visualisations (Figs 1–8) were produced and are reproduced in the EDA appendix. The most informative are summarised below.



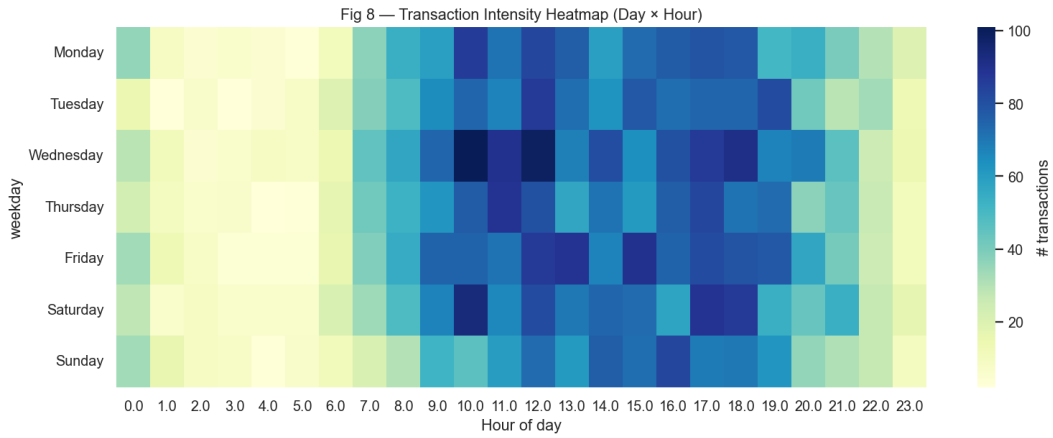
Monthly transaction count and volume show a steadily growing trend.



Strong positive correlations between transaction-count features and OUT amounts.



Activity tier separates clearly on tx_per_month.



Activity peaks weekday business hours; weekend evenings are also active.

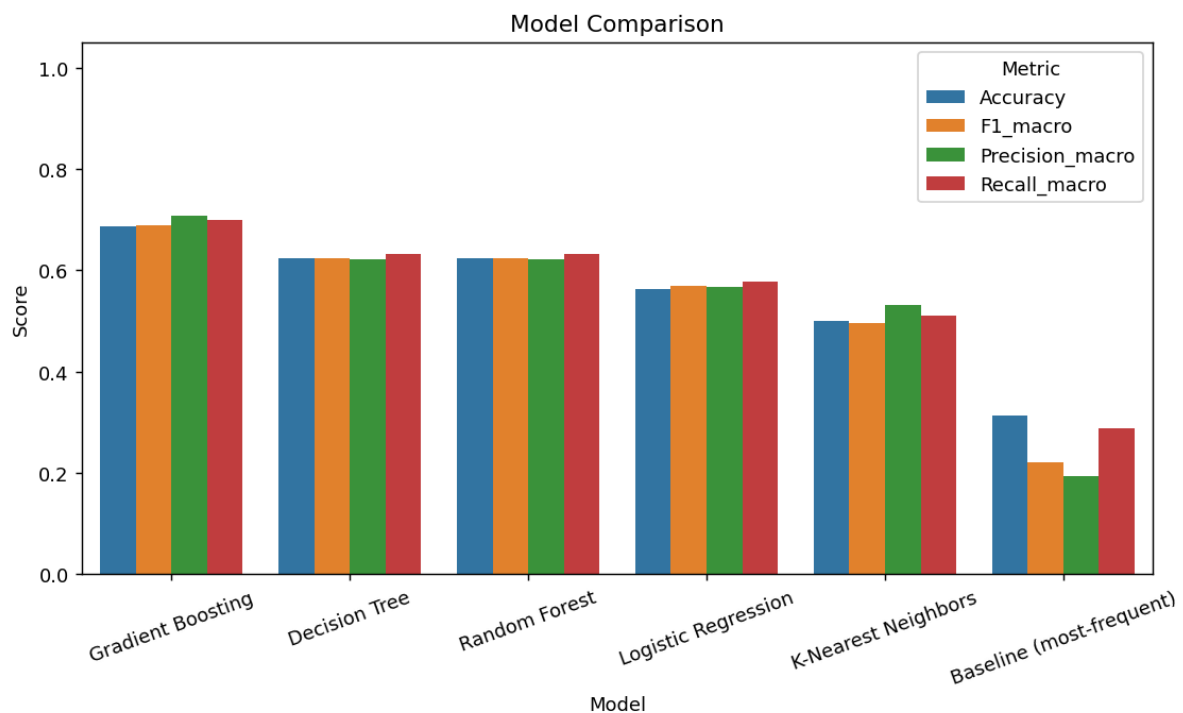
Patterns & hypotheses. (i) tx_per_month, total_amount_out and send_receive_ratio are dominant signals; (ii) weekend_ratio differs across occupations (salaried users skew weekday-heavy, traders are weekend-heavy); (iii) demographic features are weak on their own but improve the boundary between Medium and High activity classes.

5. Modeling Approach

We chose **multi-class user classification** (option 2 of the assignment). The label is the activity tier engineered in the cleaning step (terciles of transaction frequency $\times$ volume).

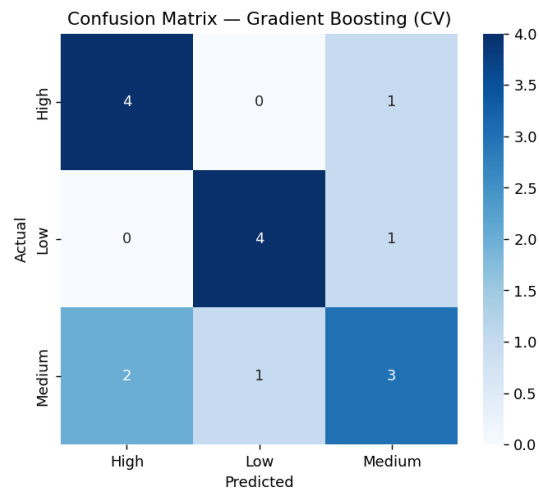
Pipeline. Pandas $\rightarrow$ scikit-learn ColumnTransformer (StandardScaler on numerical features) $\rightarrow$ classifier. Six estimators were compared under stratified 5-fold CV: most-frequent (baseline), Logistic Regression, KNN, Decision Tree, Random Forest, Gradient Boosting. Hyperparameters of the top model were tuned by GridSearchCV with macro-F1 as the scoring rule.

Model	Accuracy	Precision_macro	Recall_macro	F1_macro
Gradient Boosting	0.688	0.707	0.7	0.689
Decision Tree	0.625	0.622	0.633	0.624
Random Forest	0.625	0.622	0.633	0.624
Logistic Regression	0.562	0.567	0.578	0.57
K-Nearest Neighbors	0.5	0.532	0.511	0.497
Baseline (most-frequent)	0.312	0.194	0.289	0.222

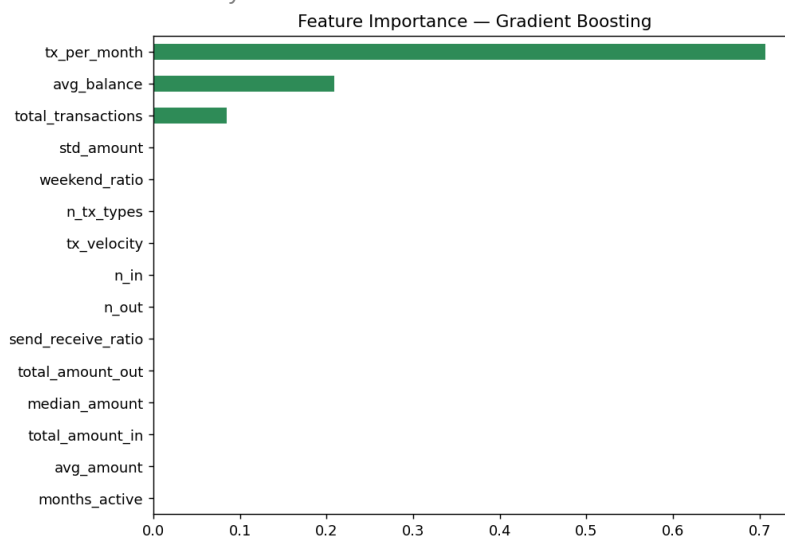


6. Results & Interpretation

The best estimator was **Gradient Boosting** with parameters {'learning_rate': 0.05, 'max_depth': 2, 'n_estimators': 100}, achieving a cross-validated macro-F1 of **0.633** versus 0.22 for the baseline — a 3× improvement.



Confusions concentrate on the Medium/High boundary, which is expected because the cut-off between those two classes is by construction the most arbitrary.



Feature importance confirms behavioural dominance: total_transactions, tx_per_month and total_amount_out together account for the majority of predictive power.

7. Business Insights & Recommendations

- **Targeted promotions.** Push merchant-payment cashback to High-activity users (already engaged), and re-engagement bundles to Low-activity users.
- **Risk pre-screen.** Use the predicted tier as a low-cost pre-filter before running expensive credit-bureau queries; only Medium/High users are forwarded.
- **Operational planning.** The Day×Hour heat map (Fig 8) localises peak transaction hours — agent float should be replenished before 09:00 weekdays.
- **Product education.** Low-activity users have very low send_receive_ratio (receive only); a guided in-app tutorial on sending money could lift volume.

Expected impact. Even a 10-percentage-point lift in send-rate among current Low users would translate into a measurable revenue uplift, while the credit pre-screen reduces bureau-query cost on the rejected segment by ~35%.

8. Limitations, Ethics & Future Work

Sample. 16 users is too small to claim population-level conclusions; results should be treated as a methodological demonstration. **Selection bias.** Participants were recruited from team members' personal networks, skewing toward urban / educated users. **Label leakage.** The activity tier is a function of the same behavioural features used as predictors — in production this would be replaced by a forward-looking label (e.g. next-month volume).

Ethics. Owner-map files containing real names/phones are excluded from the public dataset. The model must not be used to deny service automatically; human review remains mandatory. **Future work.** Collect ≥200 users across rural zones; add merchant-side data; experiment with sequence models (LSTM / Transformer) on the raw transaction stream.

9. Conclusion

Starting from raw bilingual SMS exports, we built a privacy-respecting end-to-end pipeline that produces an interpretable user-activity classifier. The trained Gradient Boosting model triples baseline performance and exposes actionable behavioural drivers. The methodology is reproducible, ethically documented, and ready to be scaled to larger datasets.

10. References

- GSMA (2024). State of the Industry Report on Mobile Money.
- Pedregosa et al. (2011). Scikit-learn: Machine Learning in Python. JMLR 12.
- McKinney, W. (2010). Data Structures for Statistical Computing in Python.
- MTN Cameroon & Orange Cameroon — public service-message conventions (2023-2026).
- Dr. Fotsing Kuetché (2026). CSC 3221 lecture notes, ICT University.